# APPLICATIONS OF ICT
# LAB 7 REPORT BE-IS(6B)

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# Introduction

This report presents the practical work completed for Applications of ICT (CS-117) at MCS, NUST. It covers computer hardware, data encoding, storage systems, number systems, hashing, and web architecture. The machine is described in Table 1 and the network trace appears in Figure 2. This report explains cybersecurity tools.

# System Specification

## Hardware and Boot Storage

This section explains the main hardware components of a computer, virtual machines, disk partitions, and the boot process. It also describes how storage devices organize data and help a computer start.

| ITEM | VALUE |
|---|---|
| **CPU model** | 13th Gen Intel(R) Core(TM) i7-1355U |
| **Physical/logical cores** | 10 physical / 12 logical |
| **RAM** | 16 GB |
| **Storage type & size** | SSD |
| **Partition style (MBR/GPT)** | GPT (GUID Partition Table) |
| **Boot mode (BIOS/UEFI)** | UEFI |
| **Windows version** | Windows 11 |
| **WSL Ubuntu version** | Ubuntu 26.04.1 LTS (WSL 2) |

*Table 1 System specification of my machine*

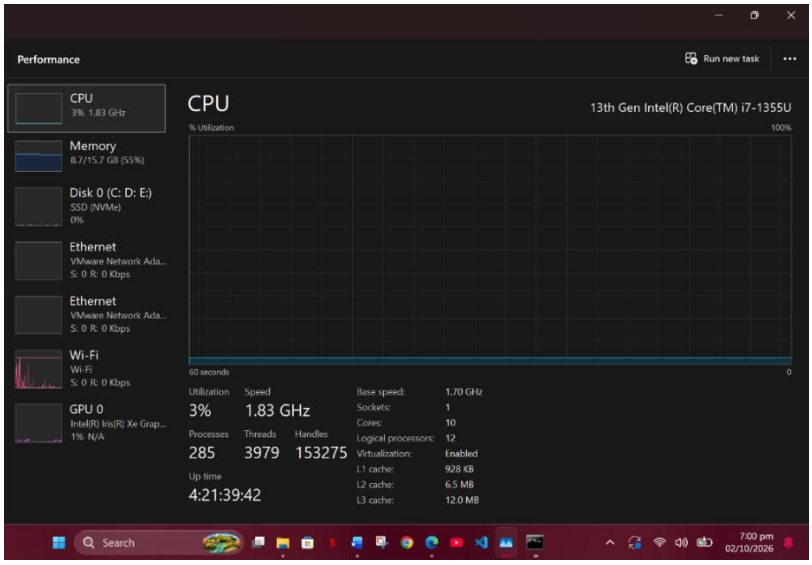



*Figure 1:System specs*

```
PS C:\WINDOWS\system32> wsl --list --verbose

  NAME      STATE           VERSION

* Ubuntu    Running         2
```

# Data and Encoding

## Number Systems and Hashing

This section explores how computers represent, process, and store information using different encoding methods. It highlights the importance of accurate data representation in digital systems.

*Table 2 Data and encoding*

| Input | CyberChef operation | Result |
|---|---|---|
| **181** | **To Binary** | 10110101 |
| **181** | **To Hex** | b5 |
| **MCS** | **To Charcode (decimal)** | 77 67 83 |
| **Alishba** | **SHA2 (256)** | 9d3d37016eb3307eb8a7f1230b3561bab8d23bc3207283a9748a67592b9bedbb |

[1]

# Web Architecture

## Network Architecture and My Web Page

This section describes the basic structure of network communication and how devices exchange information over a network. It also presents the HTML webpage developed as part of the ICT lab work.

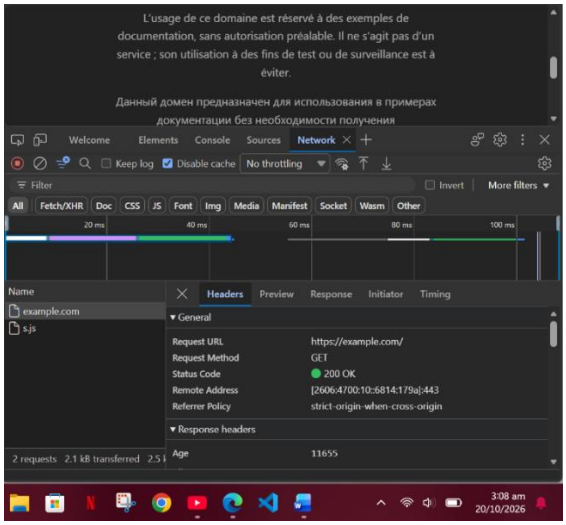



*Figure 2:Network Trace*

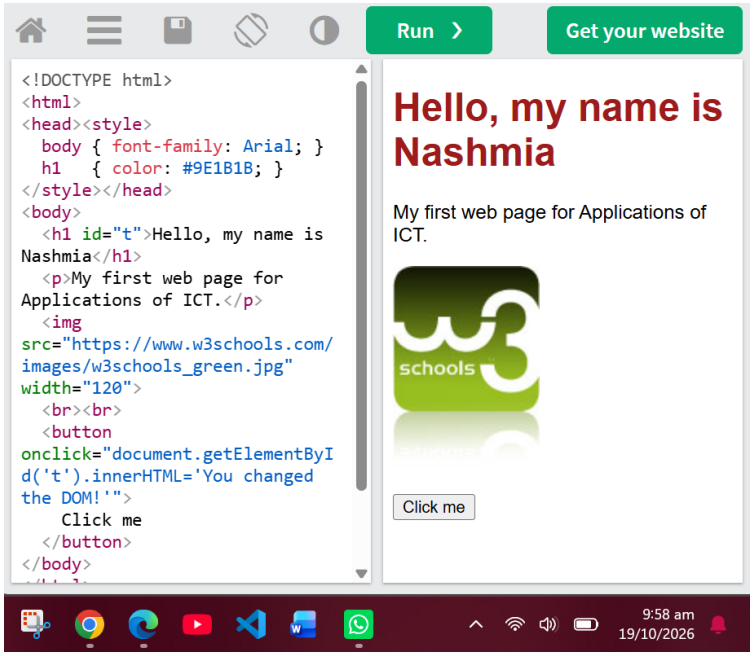



*Figure 3:Before*

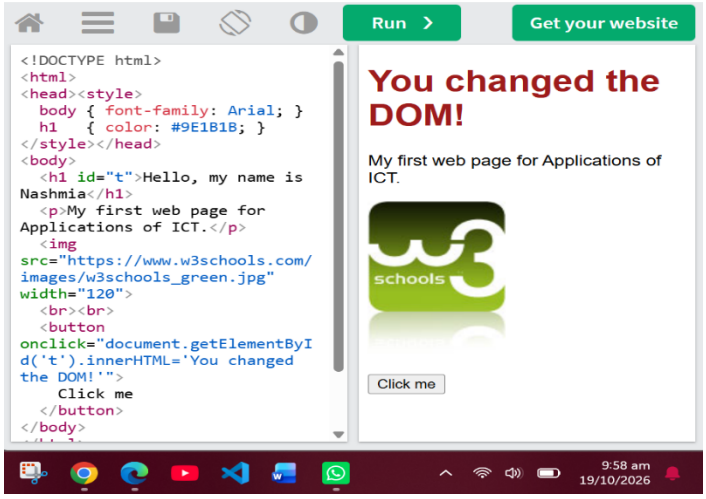



*Figure 4:After*

[2]

## Conclusion

The ICT lab activities provided practical knowledge of computer hardware, data representation, storage, networking, and web development. These exercises helped strengthen the understanding of fundamental computing concepts and their real-world applications.

[3]

# Bibliography

[1] GCHQ, "CyberChef," [Online]. Available: https://gchq.github.io/CyberChef.

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